

PMTwin PLATFORM

Collaboration Models Framework

Empowering Construction Partnerships Across Every Scale

PMTwin Platform



Need/Offer Framework

A **bidirectional marketplace** where both Needs and Offers are first-class posts. Users can post what they're seeking or what they're providing, creating a dynamic ecosystem of collaboration opportunities.



5 Main Collaboration Models 15 Collaboration Models

From simple task-based engagements to complex SPVs and strategic JVs. Each model supports both Need and Offer posts with **semantic attribute mirroring** for intelligent matching.



5 Value Exchange Modes

Go beyond cash transactions. Support **Equity, Profit-Sharing, Barter, and Hybrid** arrangements, unlocking value in resource-constrained environments.



AI-Powered Matching

Intelligent wizard guides users to the right model, then performs **bidirectional semantic matching** to connect Needs with compatible Offers based on skills, budget, timeline, and value exchange preferences.

The Need/Offer Framework

Bidirectional Marketplace

Unlike traditional platforms that separate demand (job boards) from supply (profiles), PMTwin treats both **Needs** and **Offers** as first-class collaboration posts.

- ✓ **Equal Status:** A Need post ("I'm looking for...") and an Offer post ("I can provide...") use the same collaboration models.
- ✓ **Semantic Mirroring:** Attributes are inverted, "Required Skills" becomes "Available Skills", "Budget" becomes "Rate".
- ✓ **Intelligent Matching:** The AI engine performs bidirectional matching to connect compatible Needs with Offers.

NEED

A demand,
Something I don't
Have or Own

"I am seeking..."
I demand

Need Example

"I need a structural engineer for shop drawing review. Budget: 10K SAR."

OFFER

A Give,
Something I Have or
Own

"I can provide..."

Offer Example

"I offer structural engineering services. Rate: 8-12K SAR per project."

5 Exchange Modes

PMTwin enables multiple value exchange modes beyond cash.

Cash

Standard payments with defined terms and milestones.

Structural engineer, 10K SAR in 2 milestones.

Equity

Ownership stake for contribution; includes vesting.

Join our JV: 40% equity for expertise + equipment.

Profit-Sharing

Share revenue or profit without ownership transfer.

Consortium: 60–40 profit split after costs.

Barter

Direct exchange of resources or services; no cash.

Offer office space; need structural engineering.

Hybrid

Mix cash, equity, or barter and in-kind for flexible deals.

SPV: 30% cash, 50% equity, 20% Bartering

Non-monetary exchange unlocks value where cash is scarce.

Once he/she select to Barter the offer register will be opened to select and/or add new offer.

User can link one need to multiple offers

Matching Methodology

Bidirectional Semantic Matching

Matching Models

1 One Way Matching-Simple Matching

2 Two Way Matching-Dependency

3 Group Formation

4 Circular Exchange

1- One Way Matching

Semantic Matching



User A: Need Post

Task-Based Engagement

Required Skills

Structural Engineering, AutoCAD

Budget Available

10,000 - 15,000 SAR

Deadline

March 15, 2026

Location

Riyadh, Saudi Arabia



Match



Match



Partial



Match



User B: Offer Post

Task-Based Engagement

Available Skills

Structural Engineering, AutoCAD, Revit

Rate Expected

8,000 - 12,000 SAR

Availability Start Date

March 20, 2026

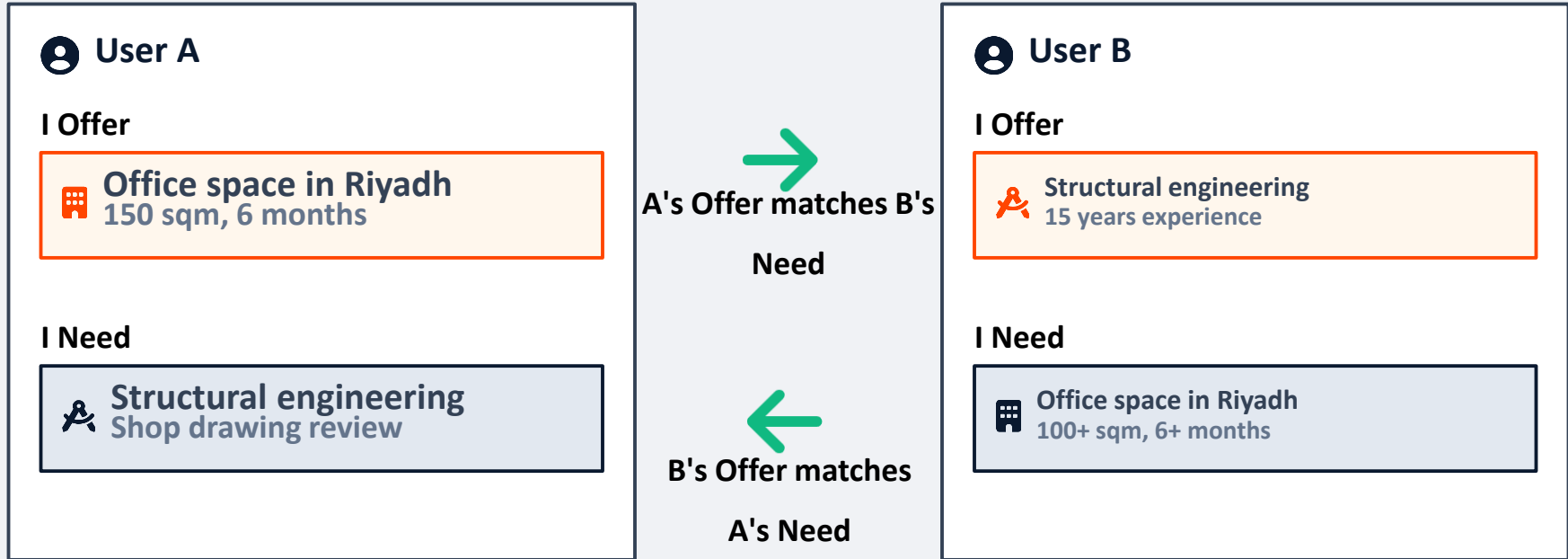
Preferred Location

Riyadh, Remote OK

MATCH SCORE: Attribute Overlap (40%) + Budget Fit (30%) + Timeline (15%) + Location (10%) + Reputation (5%) = 82%

2- Two-Way Dependency

Barter posts require **bidirectional matching**: User A's need must match User B's offer, AND User A's offer must match User B's need.



Equivalence Ratio: **1 month office space (25K SAR) = 40 hours engineering (25K SAR)**

3- Group Formation

1 Need + 3 Offers = 1 Consortium

Multiple users with **Offers** are matched with a single **Need** (or multiple Needs) to form a **single collaboration group**.

Lead Company: **Need Post**

"I need **consortium** partners for a 500M SAR highway project."

Financial:
200M SAR

Construction
Expertise

Equipment
Fleet



Matching



Partner 1: Offer

Contribution

Financial Backing

Value

200M SAR Capital



Partner 2: Offer

Contribution

Construction Expertise

Experience

20+ Highway Projects



Partner 3: Offer

Contribution

Equipment Fleet

Assets

50+ Heavy Machines



4-Party Consortium Formed

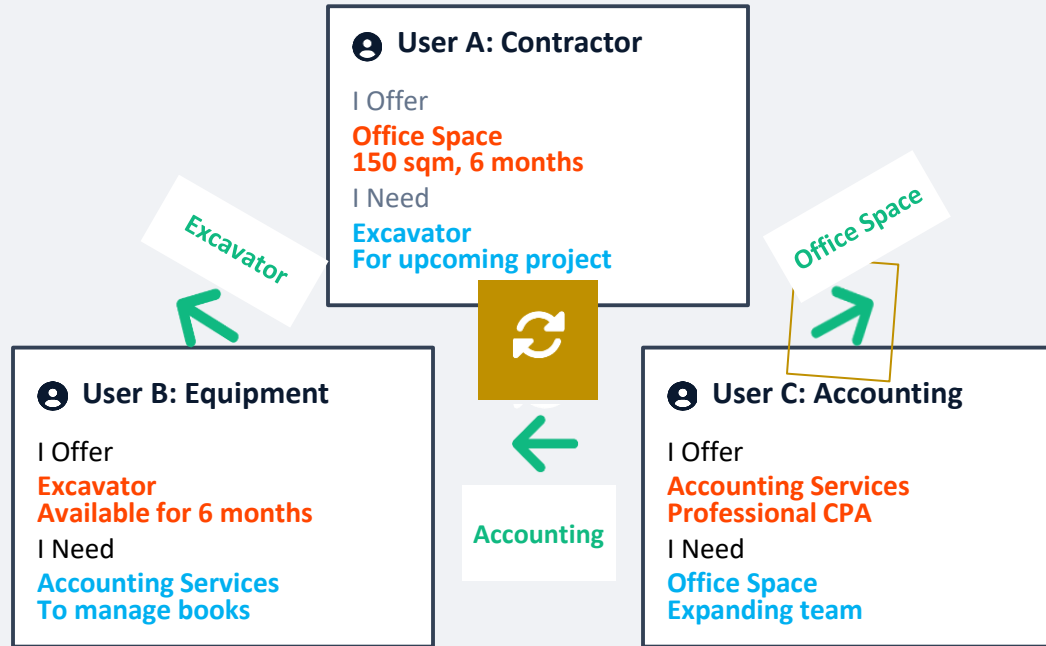
Use Cases: Consortium Formation • Bulk Purchasing • Co-Ownership Pooling • Resource Sharing Networks

4- Circular Exchange

3-Way Barter Chain

Three or more users exchange resources in a **circular pattern**, where each user's Offer matches another user's Need.

The user can link one Need to multiple Offers from his Offer register.



Zero Cash: All three parties get what they need without any money changing hands

Simplified User Journey Map

